Einstein Summation Convention

$a_1x_1 + a_2x_2 + a_3x_3 + \cdots + a_nx_n = \sum_{i=1}^{n} a_i x_i = a_i x_i$ ~ We just drop the Σ sum notation

$a_i x_i$ means

Twice repeated index MEANS sum

No index may occur more than twice in an expression

Free index shall have same range as dummy index

$a_{ii} x_k = a_{11}x_k + a_{22}x_k + \cdots + a_{nn}x_k$

$a_{ij} x_j = a_{i1}x_1 + a_{i2}x_2 + \cdots + a_{in}x_n$

row, column

$y_i = a_{ir} x_r$

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$$\left.\begin{aligned} y_1 &= a_{11}x_1 + a_{12}x_2 + a_{13}x_3 \\ y_2 &= a_{21}x_1 + a_{22}x_2 + a_{23}x_3 \\ y_3 &= a_{31}x_1 + a_{32}x_2 + a_{33}x_3 \end{aligned}\right\} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

Double Sums

$a_{ij} x_i y_j$ → n^2 terms

$i \& j \to n$

$a_{1j} x_1 y_j + a_{2j} x_2 y_j + a_{3j} x_3 y_j$

$a_{11} x_1 y_1 + a_{12} x_1 y_2 + a_{13} x_1 y_3 + a_{31} x_3 y_1 + a_{32} x_3 y_2 + a_{33} x_3 y_3$

$+ a_{21} x_2 y_1 + a_{22} x_2 y_2 + a_{23} x_2 y_3$

$y_i = c_i^r a_{rs} x_s$ let $n = 2$

$y_1 = c_1^1 a_{11} x_1 + c_1^2 a_{21} x_1 + c_1^1 a_{12} x_2 + c_1^2 a_{22} x_2$

$y_2 = c_2^1 a_{11} x_1 + c_2^2 a_{21} x_1 + c_2^1 a_{12} x_2 + c_2^2 a_{22} x_2$